

Answer to the Question No-1

(a)

$$f_{in} = \frac{\omega}{2\pi} = \frac{400\pi}{2\pi} = 200 \text{ Hz}$$

$$f_{out} = 2f_{in} = 2 \times 200 = 400 \text{ Hz}$$

(b)

$$V_s(t) = V_m \sin(\omega t)$$

$$V_s(t) = 7 \sin(400\pi t)$$

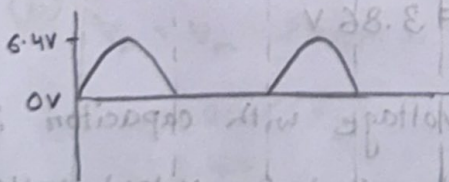
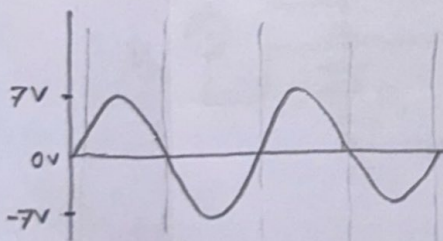
It's a sin wave fluctuating between +7V and -7V

Output,

$$V_p = V_m - 2V_{D0}$$

$$= 7 - 2(0.3)$$

$$= 6.4 \text{ V}$$



(c) Full wave without filter capacitor,

$$V_{DC} = \frac{2V_m}{\pi} - 2V_{D_0} = \frac{2 \times 7}{\pi} - 2 \times 0.3 = 3.86 \text{ V}$$

(d) With capacitor,

$$V_{\pi(P-P)} = \frac{V_p}{2fRC} = \frac{6.4}{2 \times 200 \times 5 \times 10^3 \times 100 \times 10^{-6}} = 0.032 \text{ V}$$

(e) With capacitor,

$$V_{DC} = V_p - \frac{V_{\pi(P-P)}}{2} = 6.4 - \frac{0.032}{2} = 6.384 \text{ V}$$

From 'c' without capacitor, $V'_{DC} = 3.86 \text{ V}$

$$\begin{aligned} \frac{V'_{DC}}{V_{DC}} &= \frac{3.86}{6.384} \\ \Rightarrow \frac{V'_{DC}}{V_{DC}} &= 0.606 \\ \Rightarrow V_{DC} &= 1.65 V'_{DC} \end{aligned}$$

Output voltage with capacitor is 1.65 times higher than the output voltage without capacitor. Because, capacitor charges close to peak voltage and discharges slowly through the load.

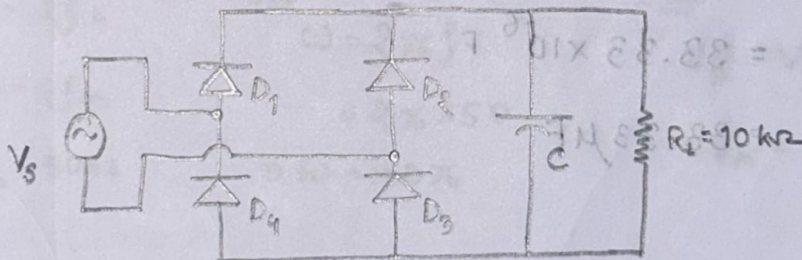
(f) To provide better filtering for the output voltage we can,

- Use capacitor with higher value.
- Use voltage regulator.
- Use RC, LC or π filter.
- Increase load resistance if possible.

(g) $f_c = 2f = 2 \times 200 = 400 \text{ Hz}$

Answer to the Question No-2

(a)



(b)
$$\begin{aligned} V_p &= V_m - 2V_{D_0} \\ &= 10 - 2 \times 0.8 \\ &= 8.4 \text{ V} \end{aligned}$$

$V_m = 10 \text{ V}$

$V_{D_0} = 0.8$

$V_{r(p-p)} = V_p \times 3\%$

$= 8.4 \times 0.03$

$= 0.252 \text{ V}$

$$(c) \quad V_{De} = V_P - \frac{V_{p(p-p)}}{2}$$

$$= 8.4 - \frac{0.252}{2}$$

$$= 8.274 \text{ V}$$

(d)

$$V_{p(p-p)} = \frac{V_P}{2fRC}$$

$$\Rightarrow C = \frac{V_P}{2fR V_{p(p-p)}}$$

$$= \frac{8.4}{2 \times 50 \times 10 \times 10^3 \times 0.252}$$

$$= 3.33 \times 10^{-5} \text{ F}$$

$$= 33.33 \times 10^{-6} \text{ F}$$

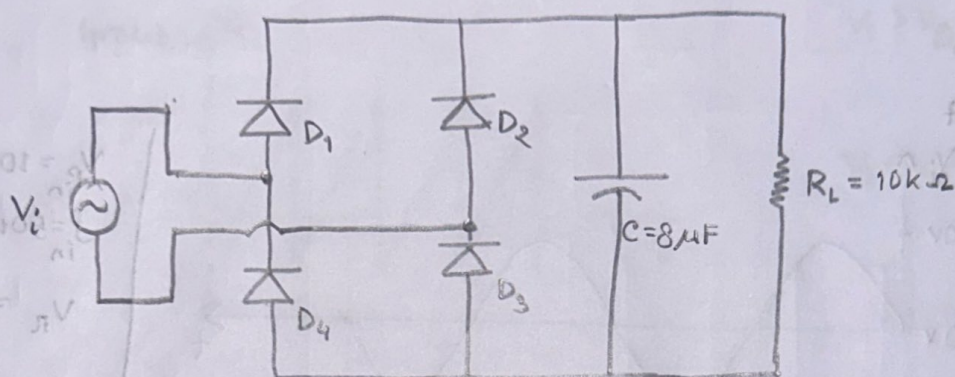
$$= 33.33 \mu\text{F}$$

$$f = \frac{\omega}{2\pi} = \frac{\omega \pi}{2\pi}$$

$$= 50 \text{ Hz}$$

Answer to the Question No-3

(a)



(b) We know,

$$V_i(t) = V_m \sin(\omega t)$$

f

$$f_{out} = 2f_{in}$$

$$\Rightarrow 100 = 2f_{in}$$

$$\Rightarrow f_{in} = 50 \text{ Hz}$$

 ω

$$\omega = 2\pi f$$

$$= 2\pi \times 50$$

$$\Rightarrow \omega = 100\pi$$

 V_m

$$V_p = V_m - 2V_{D0}$$

$$\Rightarrow V_m = V_p + 2V_{D0} \quad \text{--- (A)}$$

 V_{DC}

$$V_{DC} = I_{avg} R$$

$$= 0.75 \times 10^{-3} \times 10 \times 10^3$$

$$= 7.5 \text{ V}$$

 V_p

$$V_{DC} = V_p - \frac{V_{p(L-P)}}{2}$$

$$\Rightarrow V_{DC} = V_p - \frac{1}{2} \times \frac{V_p}{2fRC}$$

$$\Rightarrow 7.5 = V_p - \frac{1}{2} \times \frac{V_p}{2 \times 50 \times 10 \times 10^3 \times 8 \times 10^{-6}}$$

$$\Rightarrow V_p = 8 \text{ V}$$

(A) $\Rightarrow$

$$V_m = 8 + 2 \times 1$$

$$= 10 \text{ V}$$

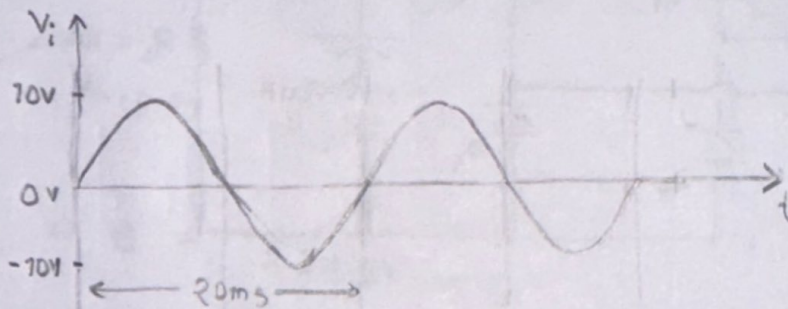
* The equation.

$$V_i(t) = 10 \sin(100\pi t) \text{ V}$$

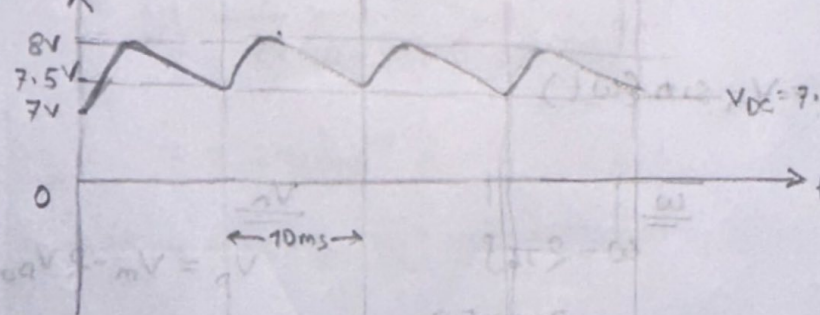
(c)

(c)

Input



Output



$$V_{P_{in}} = 10V \quad V_{P_{out}} = 8V$$

$$f_{in} = 50Hz$$

$$V_{\pi} = \frac{V_P}{8} = 1V$$

$$f_{out} = 100Hz$$

$$V_{DC} = 7.5V$$

$$T = \frac{1}{f} = \frac{1}{50} = 0.02s$$

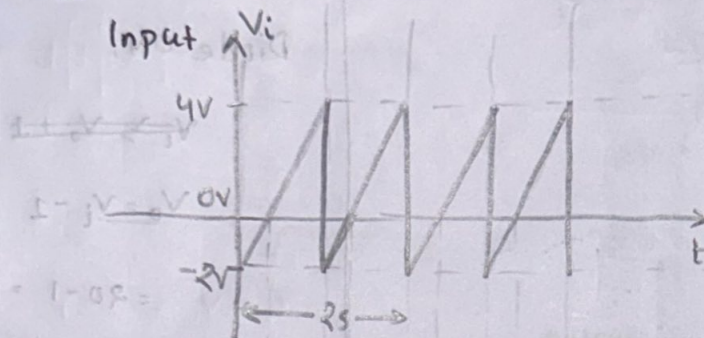
$$T_{out} = 20ms$$

$$T_{out} = \frac{1}{100} = 0.01s = 10ms$$

Answer to the Question No-4

(a)

$$\underline{\dot{V}_o} = V$$



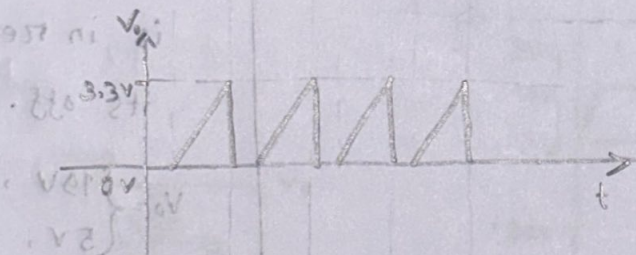
$$V_i > V_{D_0} \rightarrow \text{Diode ON}$$

$$\rightarrow V_o = V_i - 0.7$$

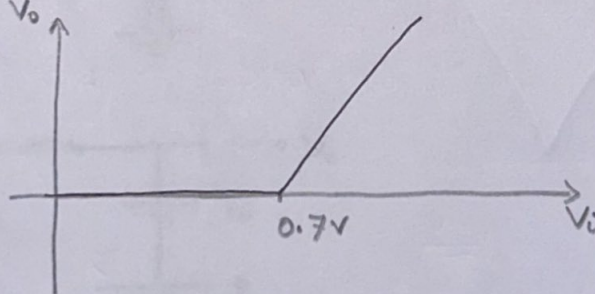
$$V_i < V_{D_0} \rightarrow \text{Diode OFF}$$

$$\rightarrow V_o = 0V$$

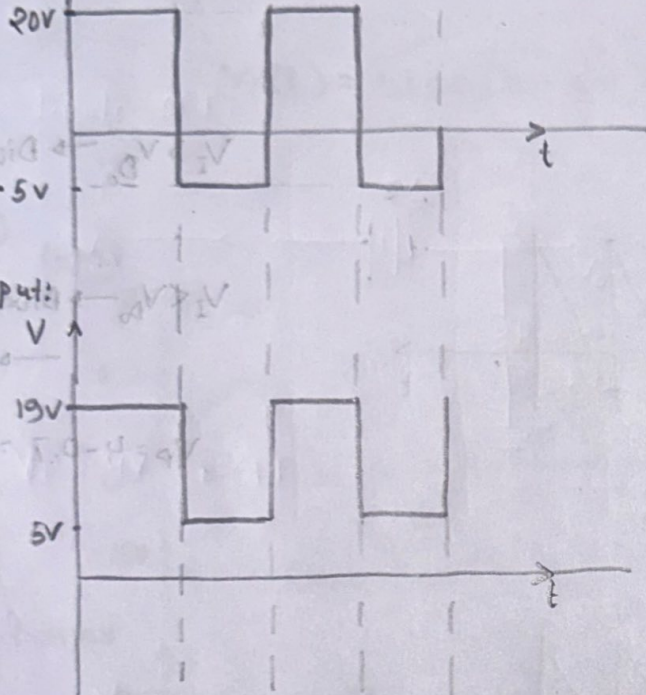
$$V_p = 4 - 0.7 = 3.3V$$



$$\underline{\dot{V}_o}$$



$$V_o = \begin{cases} 0, & \text{if } V_i < 0.7 \\ V_i - 0.7, & \text{if } V_i \geq 0.7 \end{cases}$$

(b) $V_{D0} = 1V$ 

Diode off,

No current flow,

$$V_o = 5V$$

Diode ON,

$$V_o = V_i - 1$$

$$V_o = V_i - 1$$

$$= 20 - 1 = 19V$$

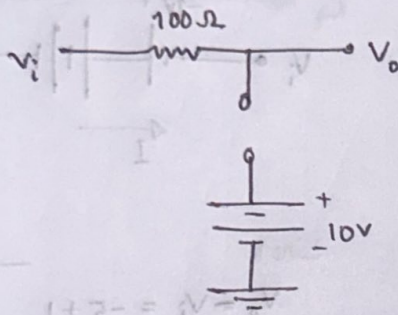
if $V_i = -5V$ then the diode is in reverse bias and its off.

$$V_o = \begin{cases} 19V, & \text{if } V_i = 20V \\ 5V, & \text{if } V_i = -5V \end{cases}$$

Answer to the Question No-5

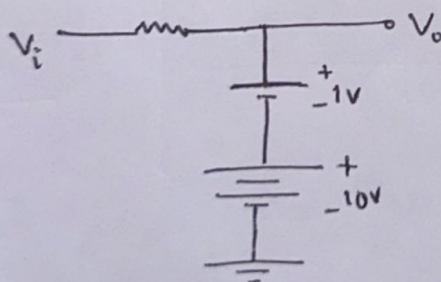
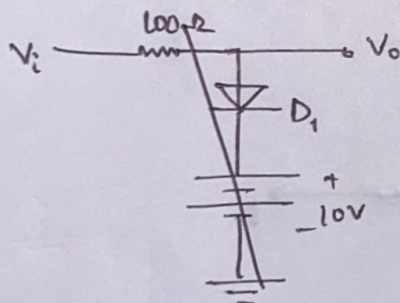
For Circuit 2, ($V_{D0} = 1V$)

Diode off.



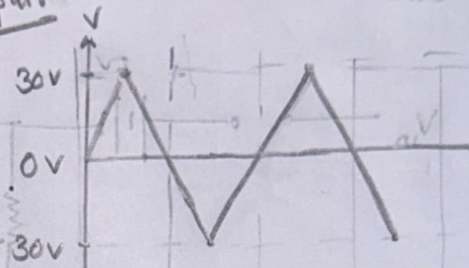
$$V_o = V_i$$

Diode ON,

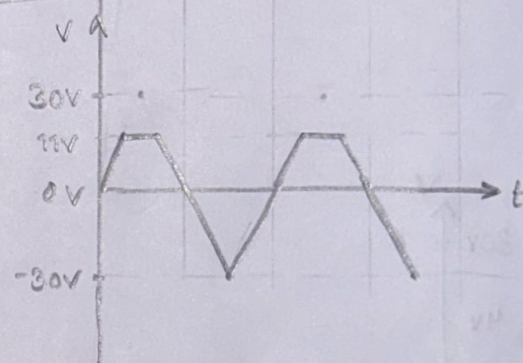


$$V_o = 1 + 10 = 11V$$

Input:

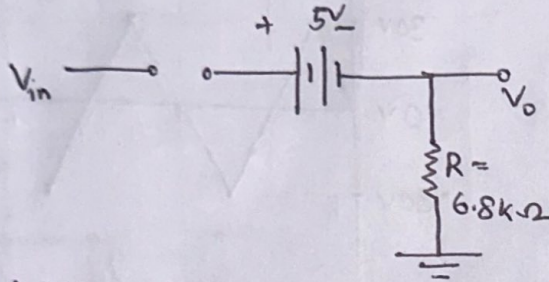


Output:

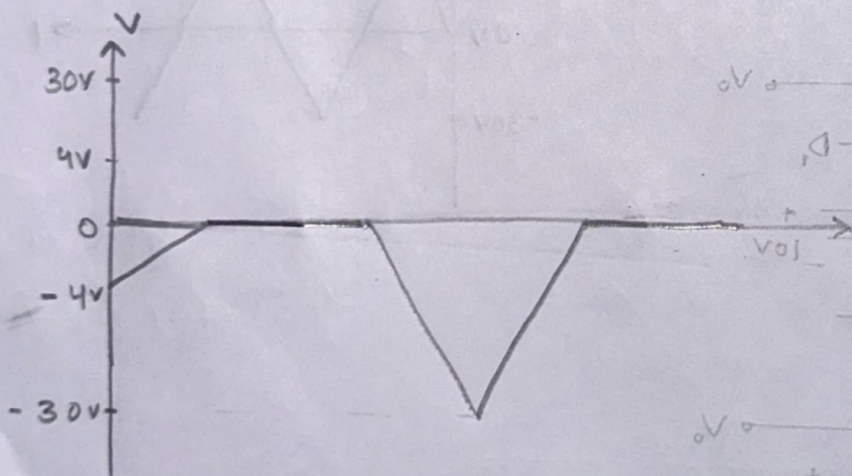


for circuit 3,

Diode OFF.

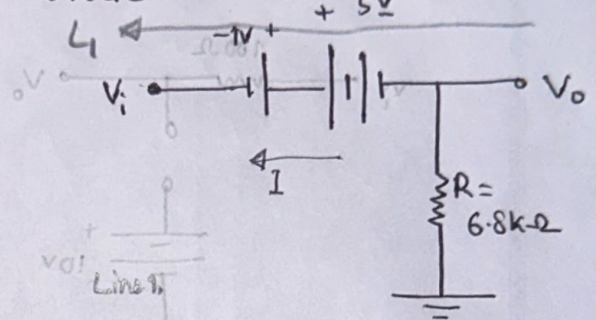


$$V_i > 4V \rightarrow \text{OFF}$$



($V_i = 0V$) & forward not

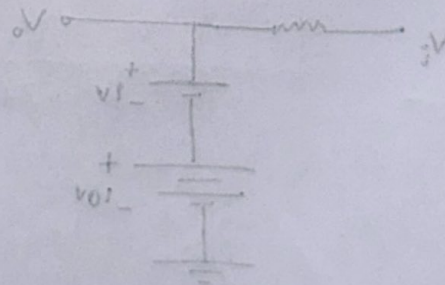
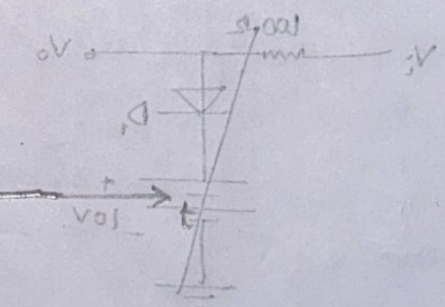
Diode ON.



$$V_o = V_i = -5 + 1$$

$$\Rightarrow V_o = V_i - 4V$$

$V_i \leq 4V \rightarrow \text{ON}$



$$V_o = 0 + 1 = 1V$$